

ARCHAI × ACMI

A local-first way for people to talk to ACMI's collection, in any language, while every record stays on hardware you own. Built to be owned, not rented.

For the Registration & Collections and Technology teams 19 August 2026 rob@fineartmedia.tech

01 The idea

We've digitised millions of objects, but most of that knowledge is still locked in cataloguing built for documentation, not for finding things or telling their stories. The quick fix on offer is commercial cloud AI. It's powerful, but it quietly moves your collection, your visitors' conversations, and your interpretive authority offshore.

ARCHAI is the other way to do it. It's an open-source layer that sits above the systems you already run. It reads from them and never writes to them, and it adds conversation, translation, rights-aware search, and a better way for staff to find things and see connections. It runs on hardware you own, so the collection, and the new layer of visitor conversation it creates, stay in the building.

WHERE THIS COMES FROM

It's my practice-based RMIT Design PhD, **Cultivating a Living Archive**. A related paper was accepted for the 6th Summit on New Media Art Archiving at ISEA2026 in Dubai. In this kind of research, building the working thing *is* the research.

02 How it works

It's a layer, not a new CMS. It connects to whatever you already run through the API, tidies the metadata, and makes the collection searchable and answerable. No migration, nothing replaced.

Three layers, kept apart on purpose

YOUR RECORDS

ACMI's own CMS / DAMS

Read-only. Nothing the AI generates ever touches the record. It stays permanent, authoritative and citable.

THE AI LAYER

Vectors + answers

Completely rebuildable from your unchanged records whenever a better model comes along. Nothing precious lives here.

EDITORIAL

Curator interpretation

Curator-written text flows one way, from archive to visitor. It never writes back.

The stack, all on your own machine

SEARCH

Qdrant + local embeddings

Rights- and place-aware, with whole-collection questions for staff.

REASONING

Local model (Qwen)

Grounded in the metadata; a five-layer check keeps answers to verified fact.

VOICE

Whisper + MeloTTS

Speech in and out, on-device, multilingual, with a genuine Australian voice.

ACCESS

Cloudflare tunnel

Locked to your domain. No collection data leaves the building.

What it can do so far

- Semantic search that doesn't rely on everyone using the same controlled words.
- Every answer keeps a trail back to the source record it came from.
- Objects that can speak in the first person, from their own verified metadata.
- A five-layer check that keeps answers grounded, even when someone tries to push them off the rails.
- A stack a museum can stand up on its own hardware and run itself.

RUNNING NOW

~1,400 public object pages

Across collections on five continents, over 8,000 staff-searchable records from 25 open sources, every object open-licensed with rights-restricted media kept out at ingest.

TO SET UP

A one-time cost

You own the box outright, and can audit it and rebuild it from scratch. It isn't a per-seat cloud subscription.

How it's built and kept honest

I build and improve it with two of my own tools. **Huggle** lets a set of AI coding agents and me work on the system together and leaves a clear, auditable record of every change. **Keytec** is a secrets and API-key wallet, so credentials stay in one controlled place and never get handed to anyone else. Keeping it improving never means loosening the sovereignty it promises.

03 Possible pilot

I'd want to start small and low-risk: a bounded set of records you approve, with the rights, terminology, cultural protocols and publication rules all set by your team. I'd build conversational pages for that set and give Collections staff a way to search across it. Your open collection API makes the connection easy, and I'd build on what you already have, including The Lens and its touch-points, rather than duplicating it. Three demos already run on real ACMI records, the green zoetrope, a Reynaud praxinoscope strip, and Muybridge's galloping horse.

SAFETY, BUILT IN FROM THE START

Public messages pass through **Saftio**, an on-device safety layer. Anyone in crisis is pointed to real human help, never an AI reply. Hate speech is held for curatorial review. Culturally sensitive material is held under your protocols. When a visitor adds a fact the curators can verify, it earns a ★ **green star** and goes into the object's record, community knowledge feeding the archive. Curators review and award that in the staff app; the public page only shows it.

04 Ways we could work together

This isn't coming out of nowhere. RMIT is already ACMI's Major University Partner, a decade-long relationship renewed in 2026, covering student placements, exhibitions and an ARC Fellowship. ARCHAI would extend that into sovereign AI research. A few ways we could go:

- ACMI as the partner site for my practice-based PhD, born-digital and screen-culture collections are exactly where this work matters most.
- An ARC Linkage project, RMIT and ACMI as partners, where your Major-Partner status genuinely strengthens the bid.
- Visitor studies, using ACMI as a place to test conversational access and the living-archive idea with real audiences.
- Working with your First Nations team so cultural protocols are built into the system itself, not just written in a policy beside it.
- Co-authored outputs, and a framework other institutions in the sector could pick up.
- Student placements building on the internship pipeline you already run.
- Grants, and, if it ever makes sense, a future deployment or licensing path for the toolkit.

05 We could explore

A few things I'd want to work out with you. This is a collaboration to shape together, not a product to install:

- ? Which slice of the collection makes the best low-risk first pilot?
- ? Which systems should it read from, and what does the open collection API expose?
- ? What rights and licensing constraints apply to the records and media in scope?
- ? What cultural protocols and First Nations governance need to hold from day one?
- ? How would you want to govern the new layer of visitor-conversation data?

- ? Is the appetite a light informal pilot, or a formal research partnership?
 - ? Could ACMI host visitor studies to see whether the living-archive idea actually works?
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06 Why ACMI

I know ACMI from the inside. I worked across Visitor Experience, Public Programs, Exhibitions, Cinema Technician and AV, and was part of the Nodel integration and the permanent-gallery update, alongside Museums Victoria, Grande Experiences and the National Communication Museum, where I was most recently Head of Technology. ACMI's born-digital, screen-culture and time-based collections are exactly where this kind of local, rights-aware interpretation and replay matters most. And the timing is real, every cloud provider now has a pitch for museums. For me, keeping it local isn't only an ethics thing, it's about being able to keep your collection playable and yours.

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Code: github.com/rob-e-graham

A concept demo and an independent proposal, not affiliated with or endorsed by ACMI.